Stochastic Neoclassical Growth Model with Labor

Model Specification and Chebyshev Polynomial Solution Method

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1 Model Specification

1.1 Household Problem

The representative household maximizes expected lifetime utility:

$$\max_{\{c_t, \ell_t, k_{t+1}\}_{t=0}^{\infty}} E_0 \sum_{t=0}^{\infty} \beta^t U(c_t, \ell_t) \tag{1}$$

subject to the resource constraint:

$$k_{t+1} = (1-\delta)k_t + z_t k_t^{\alpha} \ell_t^{1-\alpha} - c_t \tag{2}$$

where:

- c_t is consumption
- ℓ_t is labor supply (bounded: $0 \leq \ell_t \leq 1$)
- k_t is capital
- z_t is productivity (stochastic)
- $\beta \in (0,1)$ is the discount factor
- $\delta \in (0,1)$ is the depreciation rate
- $\alpha \in (0,1)$ is the capital share

1.2 Utility Function

We use log utility with labor disutility:

$$U(c, \ell) = \log(c) - \chi \frac{\ell^{1+1/\nu}}{1+1/\nu} \tag{3}$$

where:

- $\chi > 0$ is the labor disutility parameter
- $\nu > 0$ is the Frisch elasticity parameter (inverse of labor supply elasticity)

The marginal utilities are:

$$U_c(c, \ell) = \frac{1}{c} \tag{4}$$

$$U_\ell(c, \ell) = -\chi \ell^{1/\nu} \tag{5}$$

1.3 Productivity Process

Productivity follows an AR(1) process in logs:

$$\log(z_{t+1}) = \rho \log(z_t) + \varepsilon_{t+1} \tag{6}$$

where:

- $\rho \in (0, 1)$ is the persistence parameter
- $\varepsilon_{t+1} \sim N(0, \sigma^2)$ is an i.i.d. shock

2 Equilibrium Conditions

2.1 Euler Equation (Intertemporal FOC)

The Euler equation for capital accumulation is:

$$U_c(c_t, \ell_t) = \beta E_t \left[U_c(c_{t+1}, \ell_{t+1}) R_{t+1} \right] \tag{7}$$

where R_{t+1} is the return on capital:

$$R_{t+1} = \alpha z_{t+1} k_{t+1}^{\alpha-1} \ell_{t+1}^{1-\alpha} + (1 - \delta) \tag{8}$$

With log utility ($U_c(c, \ell) = 1/c$), the Euler equation becomes:

$$\frac{1}{c_t} = \beta E_t \left[\frac{1}{c_{t+1}} R_{t+1} \right] \tag{9}$$

2.2 Intratemporal FOC (Labor-Leisure Choice)

The intratemporal first-order condition equates the marginal rate of substitution between consumption and labor to the wage:

$$-\frac{U_\ell(c_t, \ell_t)}{U_c(c_t, \ell_t)} = w_t \tag{10}$$

where $w_t = (1 - \alpha) z_t k_t^\alpha \ell_t^{-\alpha}$ is the wage.
Substituting the utility derivatives:

$$\frac{\chi \ell_t^{1/\nu}}{1/c_t} = (1 - \alpha) z_t k_t^\alpha \ell_t^{-\alpha} \tag{11}$$

Simplifying:

$$\chi \ell_t^{1/\nu} = (1 - \alpha) z_t k_t^\alpha \ell_t^{-\alpha} \frac{1}{c_t} \tag{12}$$

Rearranging:

$$\chi \ell_t^{1/\nu + \alpha} = (1 - \alpha) z_t k_t^\alpha \frac{1}{c_t} \tag{13}$$

Solving for labor:

$$\ell_t = \left[\frac{(1 - \alpha) z_t k_t^\alpha}{\chi c_t} \right]^{\nu/(1+\alpha\nu)} \tag{14}$$

2.3 Resource Constraint

The resource constraint is:

$$k_{t+1} = (1-\delta)k_t + z_t k_t^{\alpha} \ell_t^{1-\alpha} - c_t \tag{15}$$

2.4 Complete Set of Equilibrium Conditions

The competitive equilibrium of the stochastic neoclassical growth model with labor is characterized by the following system of equations:

1. Euler Equation (Intertemporal Optimality):

$$\frac{1}{c_t} = \beta E_t \left[\frac{1}{c_{t+1}} \left(\alpha z_{t+1} k_{t+1}^{\alpha-1} \ell_{t+1}^{1-\alpha} + (1-\delta) \right) \right] \tag{16}$$

2. Intratemporal FOC (Labor-Leisure Choice):

$$\chi \ell_t^{1/\nu} = (1-\alpha) z_t k_t^{\alpha} \ell_t^{-\alpha} \frac{1}{c_t} \tag{17}$$

which can be solved explicitly for labor:

$$\ell_t = \left[\frac{(1-\alpha) z_t k_t^{\alpha}}{\chi c_t} \right]^{\nu/(1+\alpha\nu)} \tag{18}$$

subject to the constraint:

$$0 \leq \ell_t \leq 1 \tag{19}$$

3. Resource Constraint:

$$k_{t+1} = (1-\delta)k_t + z_t k_t^{\alpha} \ell_t^{1-\alpha} - c_t \tag{20}$$

4. Productivity Process:

$$\log(z_{t+1}) = \rho \log(z_t) + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, \sigma^2) \tag{21}$$

5. Return on Capital:

$$R_{t+1} = \alpha z_{t+1} k_{t+1}^{\alpha-1} \ell_{t+1}^{1-\alpha} + (1-\delta) \tag{22}$$

6. Wage Rate:

$$w_t = (1-\alpha) z_t k_t^{\alpha} \ell_t^{-\alpha} \tag{23}$$

The equilibrium is solved by finding policy functions $c(k_t, z_t)$ and $\ell(k_t, z_t)$ that satisfy equations (16)–(20) for all (k_t, z_t) in the state space. The policy functions are approximated using Chebyshev polynomials, and the system is solved via fixed-point iteration as described in Algorithm 4.2.

2.5 Steady State

In steady state with $z_{ss} = 1$:

$$k_{ss} = \left(\frac{\beta\alpha}{1-\beta(1-\delta)} \right)^{1/(1-\alpha)} \tag{24}$$

$$\ell_{ss} : \text{solved from intratemporal FOC} \tag{25}$$

$$c_{ss} = k_{ss}^{\alpha} \ell_{ss}^{1-\alpha} - \delta k_{ss} \tag{26}$$

3 Chebyshev Polynomial Approximation

3.1 Chebyshev Polynomials

Definition 3.1 (Chebyshev Polynomials of the First Kind)**.** Chebyshev polynomials of the first kind are defined recursively:

$$T_0(x) = 1 \tag{27}$$

$$T_1(x) = x \tag{28}$$

$$T_n(x) = 2xT_{n-1}(x) - T_{n-2}(x), \quad n \geq 2 \tag{29}$$

for $x \in [-1, 1]$. These polynomials form an orthogonal basis on $[-1, 1]$ with respect to the weight function $w(x) = 1/\sqrt{1 - x^2}$.

3.2 Chebyshev Nodes

Proposition 3.1 (Optimality of Chebyshev Nodes)**.** The Chebyshev nodes (extrema of Chebyshev polynomials) are:

$$x_i = \cos\left(\frac{(2i-1)\pi}{2n}\right), \quad i = 1, 2, \ldots, n \tag{30}$$

These nodes are optimal for polynomial interpolation in the sense of minimizing the maximum interpolation error (minimax property). They cluster near the boundaries of the interval, providing better approximation quality compared to equally-spaced nodes.

3.3 Approximation of Policy Functions

We approximate the consumption policy function $c(k, z)$ using a tensor product of Chebyshev polynomials:

$$c(k, z) \approx \sum_{i=0}^{n_k-1} \sum_{j=0}^{n_z-1} \gamma_{ij} T_i(\tilde{k}) T_j(\tilde{z}) \tag{31}$$

where:

- $\tilde{k} = \frac{2k-(k_{high}+k_{low})}{k_{high}-k_{low}}$ maps $k \in [k_{low}, k_{high}]$ to $[-1, 1]$
- $\tilde{z} = \frac{2\log(z)-(\log(z_{high})+\log(z_{low}))}{\log(z_{high})-\log(z_{low})}$ maps $\log(z)$ to $[-1, 1]$
- γ_{ij} are the Chebyshev coefficients
- n_k and n_z are the number of Chebyshev nodes for capital and productivity

3.4 Domain Selection

The capital domain is centered around steady state:

$$k_{low} = 0.5 \cdot k_{ss}, \quad k_{high} = 1.5 \cdot k_{ss} \tag{32}$$

The productivity domain covers ± 3 standard deviations:

$$\log(z_{low}) = -3\sqrt{\frac{\sigma^2}{1-\rho^2}} \tag{33}$$

$$\log(z_{high}) = 3\sqrt{\frac{\sigma^2}{1-\rho^2}} \tag{34}$$

4 Numerical Solution Method

4.1 Gauss-Hermite Quadrature

To compute expectations over productivity shocks, we use Gauss-Hermite quadrature:

$$E[f(\varepsilon)] = \int_{-\infty}^{\infty} f(\varepsilon) \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon^2}{2\sigma^2}\right) d\varepsilon \tag{35}$$

Using the change of variable $u = \varepsilon/(\sqrt{2}\sigma)$:

$$E[f(\varepsilon)] = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} f(\sqrt{2}\sigma u) \exp(-u^2) du \tag{36}$$

Gauss-Hermite quadrature approximates this as:

$$E[f(\varepsilon)] \approx \frac{1}{\sqrt{\pi}} \sum_{i=1}^{n_q} w_i f(\sqrt{2}\sigma u_i) \tag{37}$$

where (u_i, w_i) are the Gauss-Hermite nodes and weights.

4.2 Fixed-Point Iteration Algorithm

Algorithm

[title=Fixed-Point Iteration Algorithm for Consumption Policy Function]

Input: Model parameters: $\beta, \alpha, \delta, \chi, \nu, \rho, \sigma$; Chebyshev collocation nodes: (k_i, z_j) for $i = 1, \ldots, n_k$, $j = 1, \ldots, n_z$; Convergence tolerance: tol > 0; Maximum iterations: max_iter $\in \mathbb{N}$; Damping parameter: $\lambda \in (0, 1]$

Output: Chebyshev coefficients γ_{ij} defining the consumption policy function $c(k, z)$

```
// Initialization
c^(0)(k_i, z_j) ← c_ss  ∀(k_i, z_j);  γ^(0) ← 0;  iter ← 0
while iter < max_iter do
  iter ← iter + 1
  for i = 1 to n_k do
    for j = 1 to n_z do
      c_ij^(iter−1) ← c^(iter−1)(k_i, z_j)
      ℓ_ij^unc ← [ (1−α) z_j k_i^α / (χ c_ij^(iter−1)) ]^(ν/(1+αν))  ;  ℓ_ij ← min(max(ℓ_ij^unc, 0), 1)
      k'_ij ← (1 − δ) k_i + z_j k_i^α ℓ_ij^(1−α) − c_ij^(iter−1)
      E_ij ← 0
      for q = 1 to n_q do
        log(z'_qj) ← ρ log(z_j) + √2 σ u_q ;  z'_qj ← exp(log(z'_qj))
        c'_qj ← c^(iter−1)(k'_ij, z'_qj)
        ℓ'_qj ← [ (1−α) z'_qj (k'_ij)^α / (χ c'_qj) ]^(ν/(1+αν))  ;  ℓ'_qj ← min(max(ℓ'_qj, 0), 1)
        R'_qj ← α z'_qj (k'_ij)^(α−1) (ℓ'_qj)^(1−α) + (1 − δ)
        E_ij ← E_ij + β (w_q / √π) (R'_qj / c'_qj)
      end for
      c_ij^new ← 1/E_ij ;  c_ij^(iter) ← (1 − λ) c_ij^(iter−1) + λ c_ij^new
      e_ij ← E_ij − 1/c_ij^(iter−1)
    end for
  end for
  γ^(iter) ← T^(−1) c^(iter)
  max_error ← max_{i,j} |e_ij|
  if max_error < tol then return γ^(iter)
end while
return γ^(iter)
```

4.3 Inversion Step

Given consumption values $c(k_i, z_j)$ at collocation points, we solve for coefficients γ_{ij} such that:

$$c(k_i, z_j) = \sum_{i'=0}^{n_k-1} \sum_{j'=0}^{n_z-1} \gamma_{i'j'} T_{i'}(\tilde{k}_i) T_{j'}(\tilde{z}_j) \tag{38}$$

This is a linear system:

$$\mathbf{c} = \mathbf{T}\boldsymbol{\gamma} \tag{39}$$

where:

- $\mathbf{c}$ is the vector of consumption values (length $n_k \times n_z$)
- $\mathbf{T}$ is the matrix of Chebyshev polynomials evaluated at collocation points
- $\boldsymbol{\gamma}$ is the vector of coefficients (length $n_k \times n_z$)

The solution is:

$$\boldsymbol{\gamma} = \mathbf{T}^{-1}\mathbf{c} \tag{40}$$

The matrix $\mathbf{T}$ is well-conditioned when using Chebyshev nodes, ensuring numerical stability.

4.4 Convergence Criterion

The algorithm converges when:

$$\max_{i,j} |e_{ij}| < \text{tol} \tag{41}$$

where e_{ij} is the Euler error at collocation point (k_i, z_j):

$$e_{ij} = \beta E_t \left[\frac{1}{c(k'_{ij}, z'_{ij})} R'_{ij} \right] - \frac{1}{c(k_i, z_j)} \tag{42}$$

5 Implementation Details

5.1 Labor Constraint

Labor is constrained to $[0, 1]$. When computing labor from the intratemporal FOC:

$$\ell_{\text{unconstrained}} = \left[\frac{(1-\alpha) z k^\alpha}{\chi c} \right]^{\nu/(1+\alpha\nu)} \tag{43}$$

we apply the constraint:

$$\ell = \min(\max(\ell_{\text{unconstrained}}, 0), 1) \tag{44}$$

5.2 Damping

To improve convergence, we use dampening:

$$c^{(iter)} = (1-\lambda) c^{(iter-1)} + \lambda c_{\text{new}}^{(iter)} \tag{45}$$

where $\lambda \in (0, 1]$ is the dampening parameter.

5.3 Error Metrics

We track two error metrics:

$$\text{Euler error} = \beta E_t \left[\frac{1}{c'} R' \right] - \frac{1}{c} \tag{46}$$

$$\text{Intratemporal error} = \chi \ell^{1/\nu} - \frac{(1-\alpha) z k^{\alpha} \ell^{-\alpha}}{c} \tag{47}$$

6 Model Parameters

Typical parameter values used in the numerical solution:

$$\beta = 0.99 \quad \text{(discount factor)} \tag{48}$$

$$\alpha = 0.33 \quad \text{(capital share)} \tag{49}$$

$$\delta = 0.025 \quad \text{(depreciation rate)} \tag{50}$$

$$\chi = 8.042775 \quad \text{(labor disutility)} \tag{51}$$

$$\nu = 1.0 \quad \text{(Frisch elasticity)} \tag{52}$$

$$\rho = 0.95 \quad \text{(productivity persistence)} \tag{53}$$

$$\sigma = 0.007 \quad \text{(productivity shock std. dev.)} \tag{54}$$

$$n_k = 10 \quad \text{(Chebyshev nodes for capital)} \tag{55}$$

$$n_z = 10 \quad \text{(Chebyshev nodes for productivity)} \tag{56}$$

$$n_q = 5 \quad \text{(Gauss-Hermite quadrature nodes)} \tag{57}$$